Lab 02 Introduction to Java programming language

Task 1: Read a string and compress consecutive characters into `char + count`.
Example: `aaabbc` → `a3b2c1 or AAAAaa` → `A4a2`
Input: one line **Output:** encoded string **Constraints:** preserve case and symbols

Task 2: Find the longest substring that is a palindrome.
Example: `babad` → `bab` (or `aba`)
Input: one line **Output:** longest palindrome substring + its length
Constraints: brute force allowed; no arrays

Task 3: Read a message and an integer shift `k`. Encrypt letters only:

- `A-Z` stays uppercase, `a-z` stays lowercase
- wrap around (e.g., `z` with shift 2 → `b`)
- non-letters unchanged
- Example : `abc k = 1` → `bcd`

Input: line + integer `k** **Output:** encrypted line **Constraints:** normalize `k` to `[0,25]`

Task 4: Read a line and output each word on a new line, ignoring extra spaces.
Example: `" I love Java "` →

```
I
love
Java
```

Input: one line **Output:** one word per line + total word count **Constraints:** no `split()`

Task 5: Read a string and check if parentheses `(` and `)` are balanced.

- `(a+(b))` → `BALANCED`
- `()))(` → `NOT BALANCED`

Input: one line **Output:** `BALANCED` or `NOT BALANCED`
Constraints: only parentheses matter; other chars ignored

Task: Read a password and print `ACCEPTED` or `REJECTED` based on these rules:
Rules (all must be satisfied)

1. Length is **between 8 and 16** (inclusive)
2. Contains **at least one uppercase** letter (`A-Z`)
3. Contains **at least one lowercase** letter (`a-z`)
4. Contains **at least one digit** (`0-9`)
5. Contains **at least one special character** from: `@ # $ % !`
6. Must **not** contain a space
7. Must **not** contain the same character repeated **3 times in a row**
   (e.g., `aaA1@bbb` is rejected because `bbb`)

Input: one line `password`
Output: `ACCEPTED` if valid
otherwise `REJECTED: <reason>` (print the first rule that fails)

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